

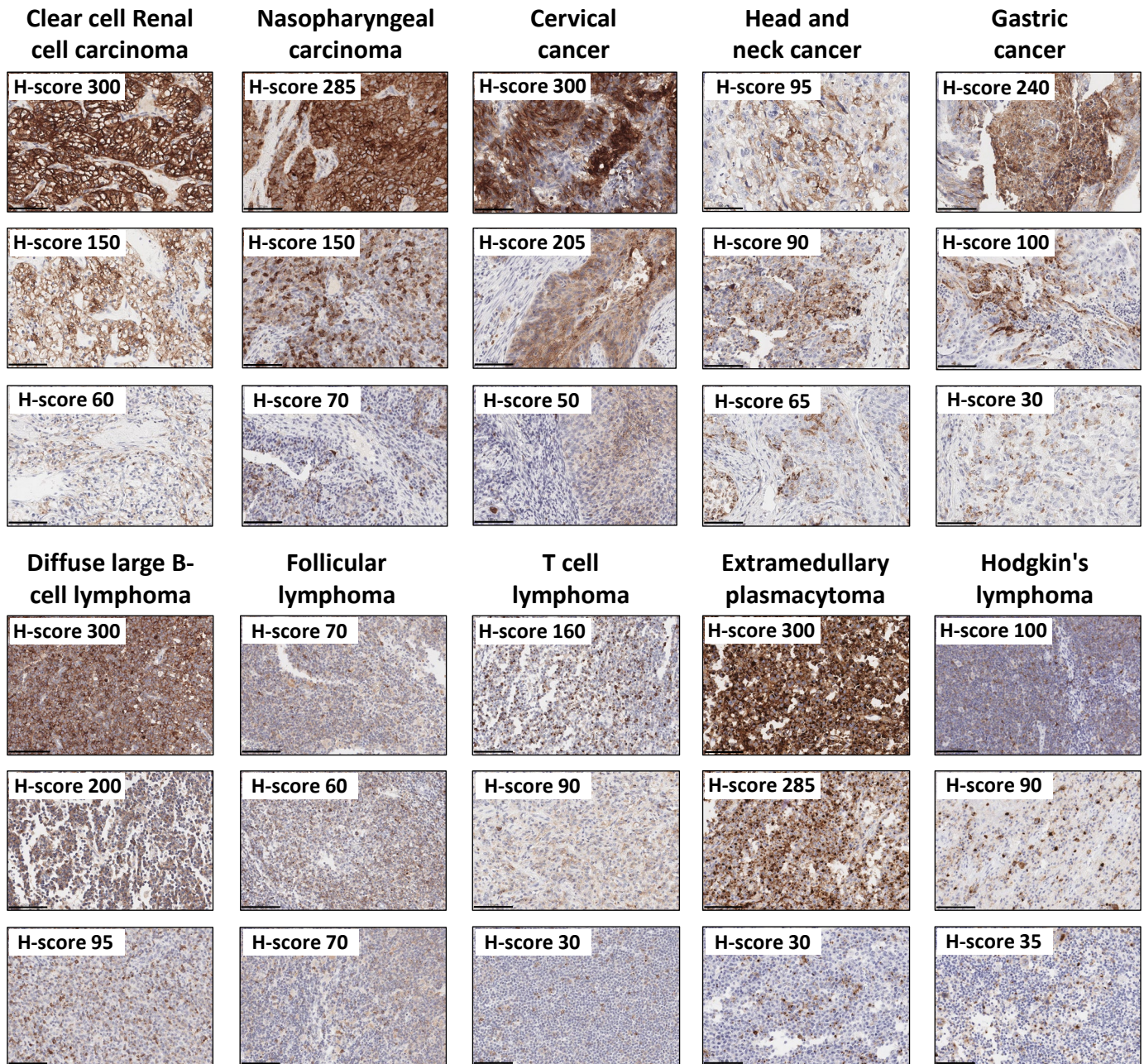
Supplemental Information for

ADI-270: an armored allogeneic gamma delta T cell therapy designed to target CD70-expressing solid and hematologic malignancies

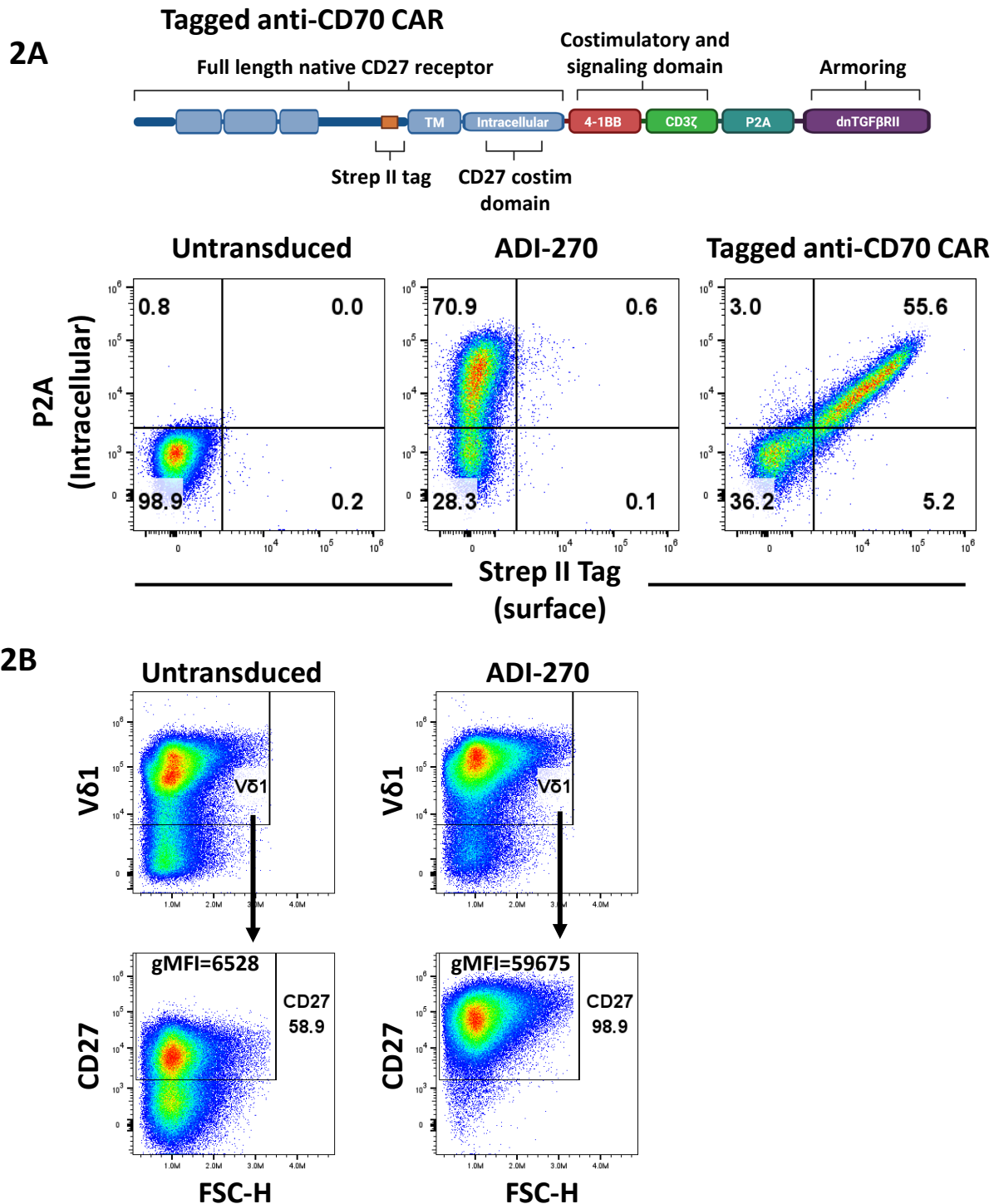
Kevin P. Nishimoto, Gauri Lamture, Yvan Chanthery, Alexander G. Teague, Yogendra Verma, Melinda Au, Morgan Smith-Boeck, Michael Salum, Pranav Murthy, Smitha R.Y. Gundurao, Ramandeep Kaur, Jie Zhang, Aruna Azameera, Jonathan T. S. Wong, Elizabeth B. Speltz, Katherine M. Wang, Amy Doan, Jyothi Sethuraman, Dishant Bhatwala, Ana Giner-Rubio, Pavan Puligujja, Helen Budworth, Christopher J. Rold, Swapna Panuganti, Aya Jakobovits, Marissa Herrman, Arun Bhat, Shon Green, Blake T. Aftab

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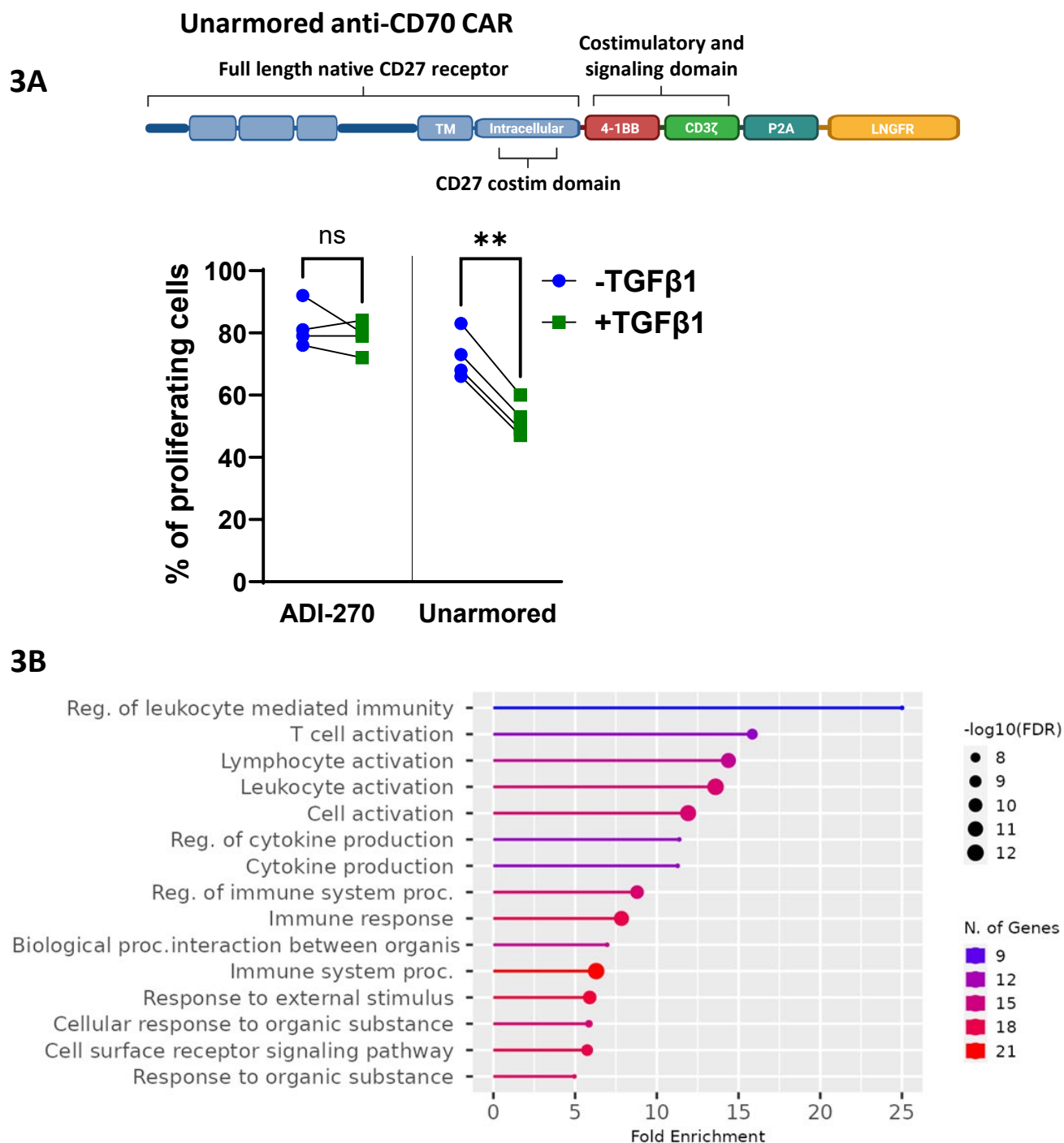
Supplemental Fig. 1 to 8, and Supplemental Table 1 and 2.



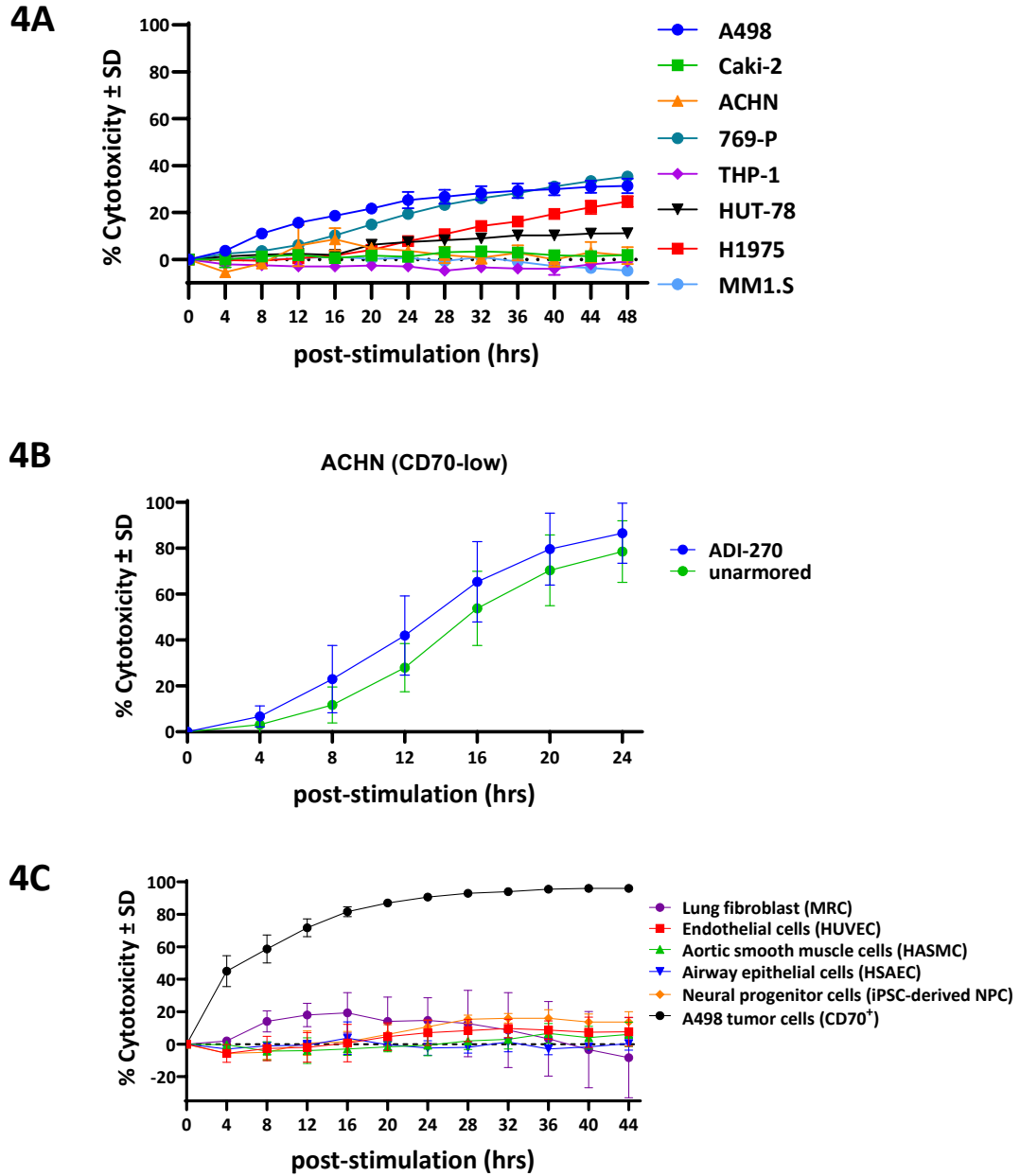
Supplemental Fig. 1. IHC staining for CD70 in patient tumor samples. Representative IHC images ($\times 20$ magnification) of CD70 staining intensities and H-scores of selected primary solid tumors and hematologic malignancies from tumor microarrays (TissueArray.com). H-scores were determined by a certified pathologist (Histowiz). The scale bar (black line) represents 100 μm . IHC, immunohistochemistry.



Supplemental Fig. 2. ADI-270 stably expresses the CD27 receptor CAR. (A) Streptavidin tag (Strep II tag), consisting of an eight amino acid sequence (WSHPQFEK), was inserted into the CAR extracellular juxtamembrane region to enable cell surface detection (Tagged anti-CD70 CAR). Untransduced, ADI-270, and tagged anti-CD70 CAR Vδ1 T cells (Strep II tag in CAR) were surface stained with anti-Strep II antibody conjugated to biotin and detected using Streptavidin-PE. Cells were treated with fixation and permeabilization buffer and stained with anti-P2A antibody. The P2A and Strep II tag-positive cell percentages were analyzed using flow cytometry. Values in the quadrant gating represent the percentage of cells in the upper left gate (P2A⁺/Strep II Tag^{negative}), lower left gate (P2A^{negative}/Strep II Tag^{negative}), upper right gate (P2A⁺/Strep II Tag⁺), and lower right gate (P2A^{negative}/Strep II Tag⁺). The data are representative of three independent experiments. **(B)** ADI-270 and untransduced Vδ1 T cells (matched donors) were stained for CD27 expression and analyzed using flow cytometry. The percentage of cell surface CD27 was derived from the gated Vδ1 T cell population. The gMFI values were calculated from the gated CD27 cell population. Data are representative of three different donors.



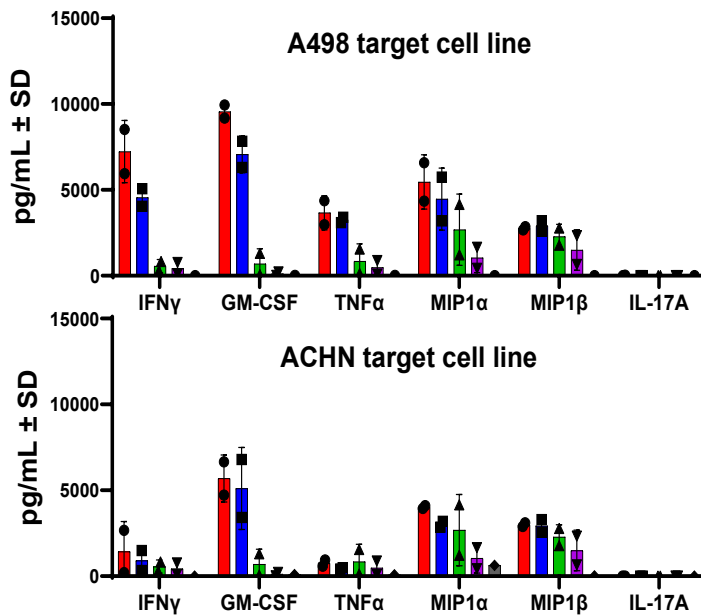
Supplemental Fig. 3. dnTGFβRII armoring protects ADI-270 from the immunosuppressive effects of TGFβ. (A) Schematic of the unarmored anti-CD70 CAR. CTV-labeled ADI-270 and unarmored anti-CD70 Vδ1 T cells (without dnTGFβRII) were stimulated with A498 tumors cells for 7 days in the presence or absence of TGFβ1 (20 ng/mL). Using flow cytometry, proliferating cells with decreased CTV expression from division zero, were determined and compared between conditions in the presence (green) and absence (blue) of TGFβ1. Percent proliferating cell values are from four different donors (symbols). Two-way ANOVA, **p<0.01. (B) Displayed are the biological pathways associated with downregulated DEGs from unarmored anti-CD70 Vδ1 T cells activated with CD70 protein in the presence of TGFβ1. Gene ontology and pathway analysis of the downregulated DEGs identified in unarmored anti-CD70 Vδ1 T cells when exposed to exogenously added TGFβ1, compared to conditions without TGFβ1 post-CD70 protein stimulation. The analysis utilized ShinyGO 0.81 (<http://bioinformatics.sdstate.edu/go/>). The circle size represents the false discovery rate (FDR), and the color code represents the number of genes in each pathway. ANOVA, analysis of variance; CTV, Cell Trace Violet; DEGs, differentially expressed genes.



Supplemental Fig. 4. ADI-270 exhibits specific anti-tumor activity in vitro. (A) The cytotoxicity of unengineered V δ 1 T cells was measured against a panel of cells derived from hematologic and solid tumors at a 1:1 E:T ratio for 48 hrs using the Incucyte SX5 instrument. Percent cytotoxicity (mean $\pm$ SD) is derived from three biological replicates. Data are representative of two different independent experiments. **(B)** The activity against CD70⁺ tumor cells was independent of dnTGF β RII armoring. The cytotoxic potential of ADI-270 and unarmored anti-CD70 CAR V δ 1 T cells was measured against the CD70-low expressing ACHN cell line at a 1:1 E:T ratio for 24 hrs using the Incucyte SX5 instrument. Percent cytotoxicity values (mean $\pm$ SD) are from four different donors. **(C)** In vitro cytotoxicity of ADI-270 against CD70^{negative} normal cells (MRC, HUVEC, HASMC, HSAEC from ATCC, and NPC from STEMCELL Tech) at a 1:1 E:T ratio was imaged every 4 hrs using the Incucyte. The Incucyte[®] Analysis Software was used to apply Cell-by-Cell analysis (Sartorius) to identify viable target adherent cells. Percent cytotoxicity values (mean $\pm$ SD) are from three different donors with the specific target cell type indicated. E:T, effector-to-target.

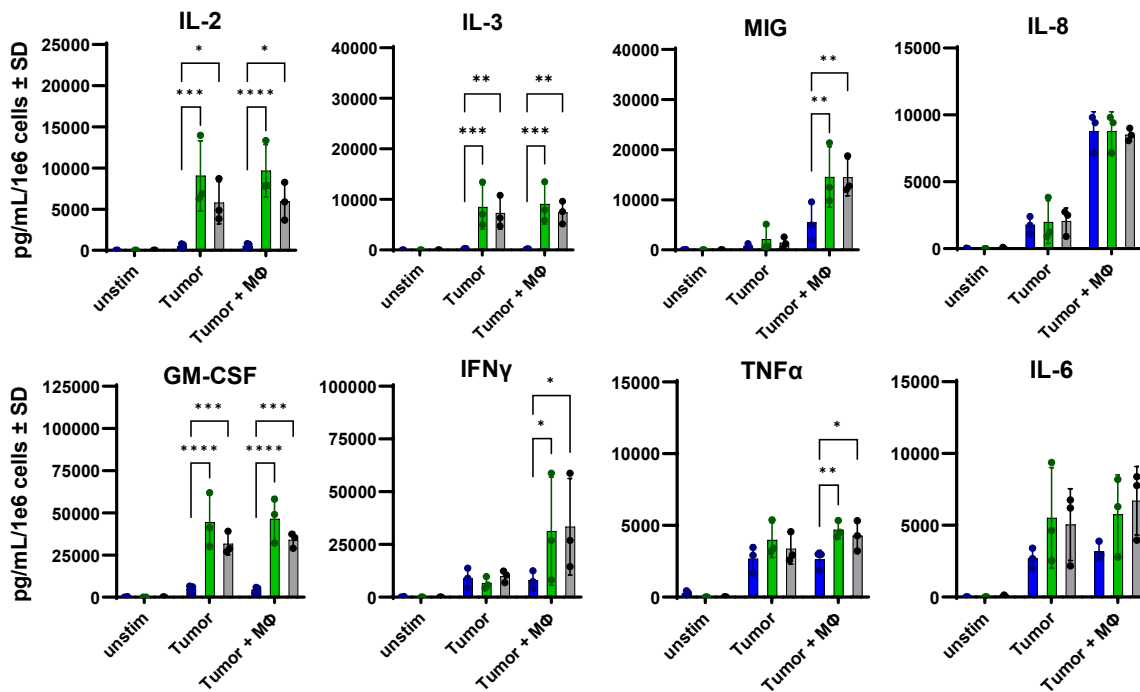
5A

ADI-270 + Target cell line ADI-270 alone Target alone
Unarmored anti-CD70 CAR + Target cell line Unarmored anti-CD70 CAR alone

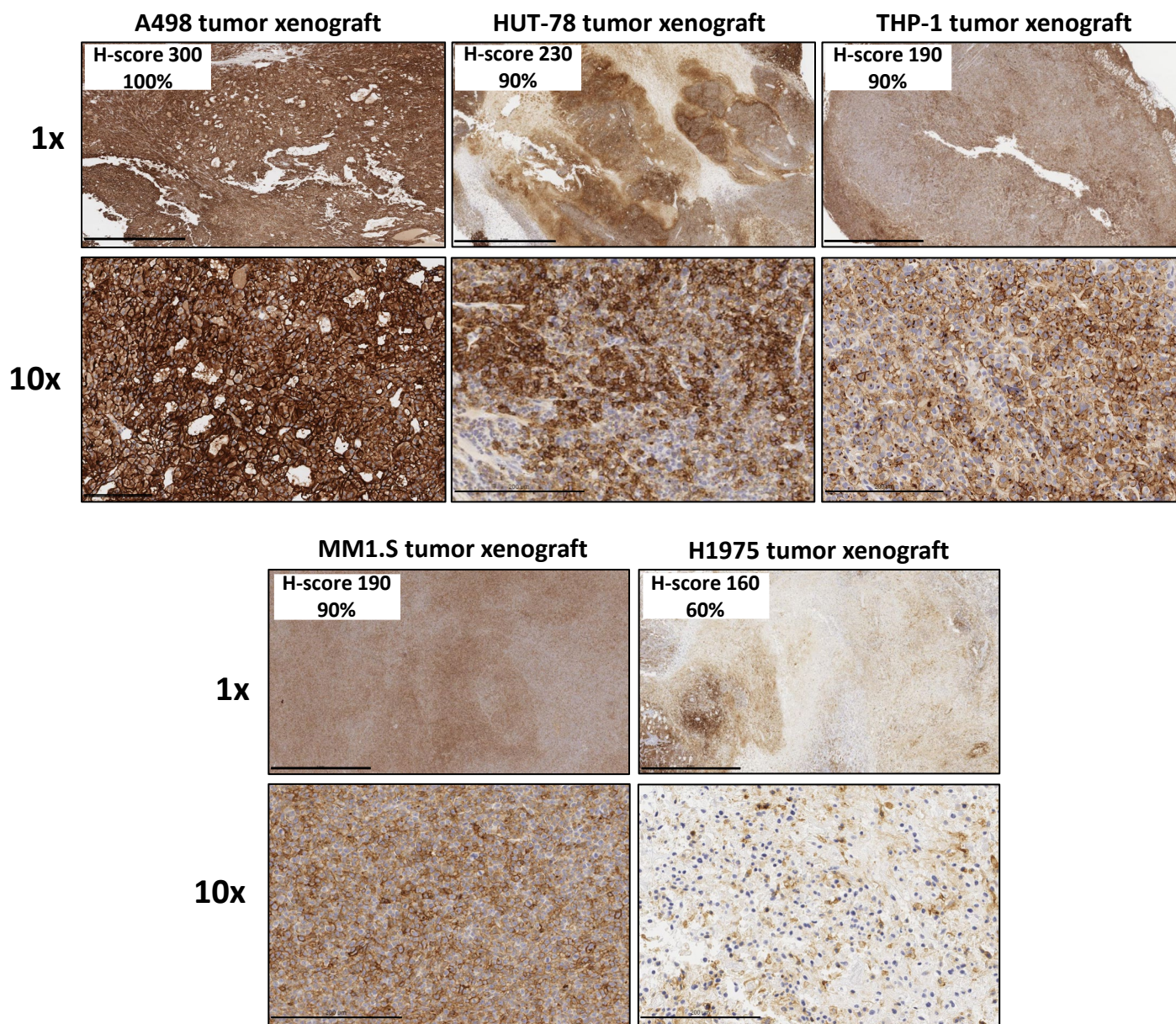


5B

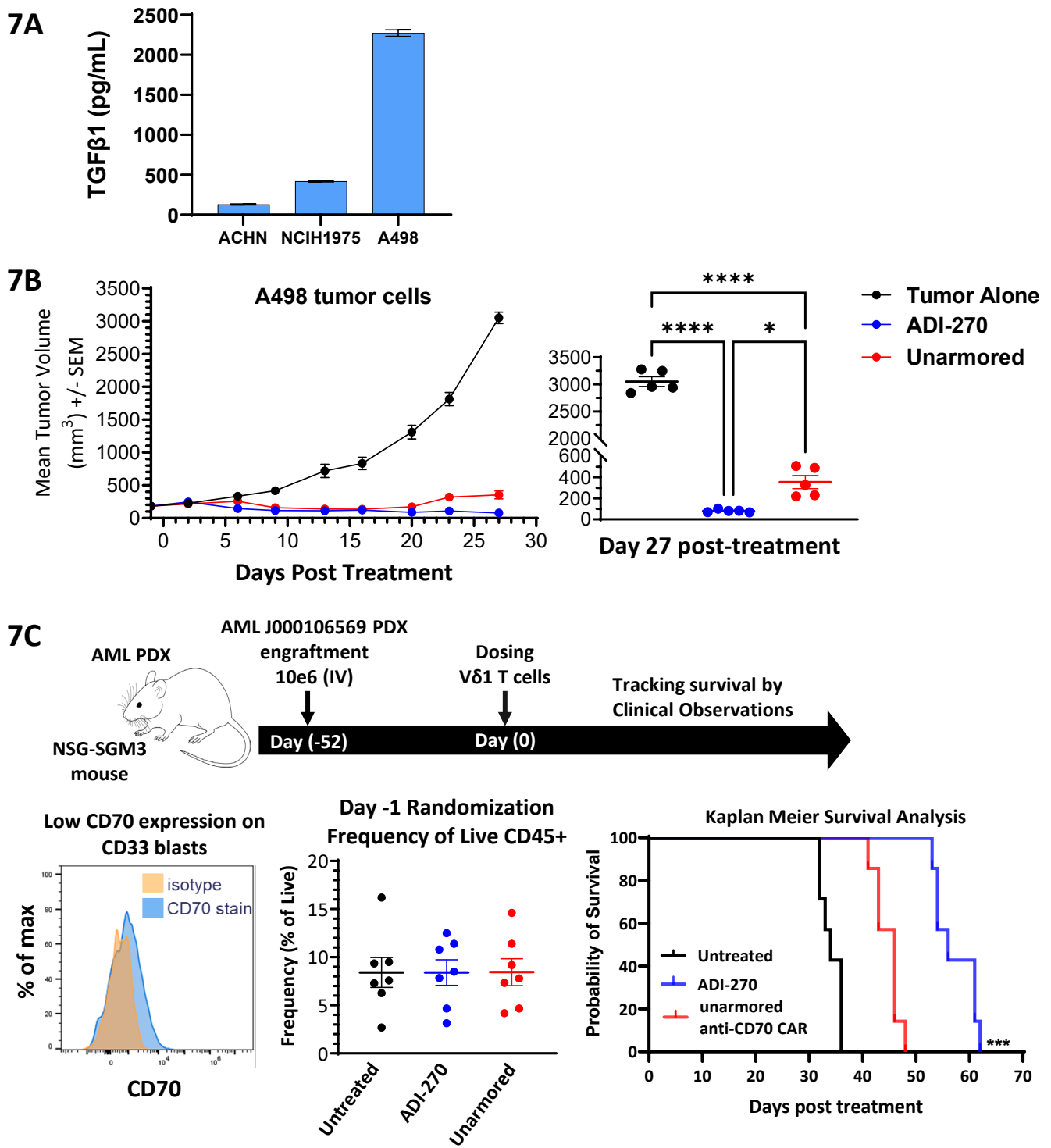
ADI-270 αβ CAR-T Ref1 αβ CAR-T Ref2



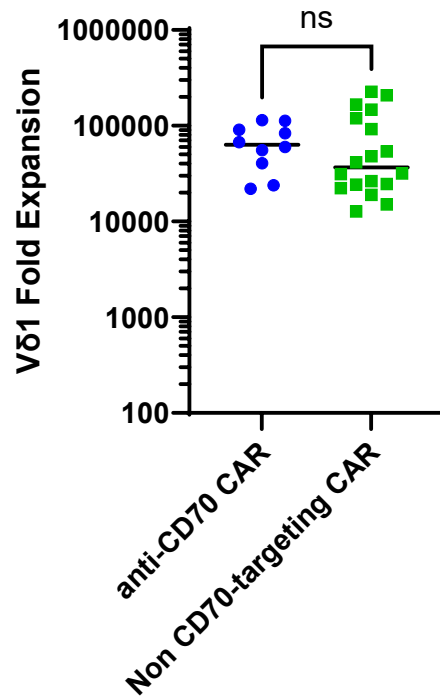
Supplemental Fig. 5. ADI-270 is associated with a favorable cytokine and chemokine profile. (A) Cytokine and chemokine production from ADI-270 and unarmored anti-CD70 CAR Vδ1 T cells after a 24-hr coculture with tumor cell lines A498 and ACHN. Levels of IFN γ , GM-CSF, TNF α , MIP1 α , MIP1 β , and IL-17A were measured in culture supernatants using the Luminex platform. Bar graphs represent the mean concentration (pg/mL $\pm$ SD) from two different donors (symbols). **(B)** Cytokines and chemokine levels from ADI-270 or anti-CD70 CAR $\alpha\beta$ T cell references cocultured with A498 tumor cells (1:1 E:T), tricultured with A498 tumor cells and macrophages (M Φ) (1:1:0.05 E:T:M Φ), or CAR T cell effector alone. M Φ s were generated by CD14 microbead positive selection (Miltenyi) from human healthy donor PBMCs and cultured in RPMI 1640 + 10% FBS in the presence of recombinant human M-CSF (50 ng/mL, R&D Systems) for 6 days. Cytokines and chemokines were measured from culture supernatant collected after 24 hrs using the Luminex platform. Data shown are the mean concentration (pg/mL/1e6 cells $\pm$ SD) using CAR T cell effectors derived from three different donors (symbols). Two-way ANOVA, *P<0.05, **P<0.01, ***P<0.001, ****P<0.0001. ANOVA, analysis of variance; E:T, effector-to-target; M-CSF, macrophage colony-stimulating factor; PBMCs, peripheral blood mononuclear cells.



Supplemental Fig. 6. CD70 is expressed on tumor xenografts from multiple solid and hematologic human tumor cell lines. Tumor xenografts from untreated NSG mice at end of study were harvested and fixed in 10% neutral buffered formalin for 48 hrs, and then preserved in 70% ethanol before being embedded in paraffin. The tumor blocks were sectioned and processed for IHC staining of CD70 (Histowiz). A certified pathologist (HistoWiz) evaluated the tumor sections for the prevalence and intensity of CD70 staining (H-score, % positive). The representative images are from tumor xenografts A498, HUT-78, THP-1, MM1.S, and H1975. The scale bar (black line) represents 2 mm and 200 μ m for the $\times 1$ and $\times 10$ magnification, respectively. IHC, immunohistochemistry.



Supplemental Fig. 7. dnTGFBR1I armoring of ADI-270 is associated with greater in vivo tumor control compared to unarmored anti-CD70 CAR Vδ1 T cells. (A) Tumor cell lines were cultured for 24 hrs in OptiMEM (ThermoFisher) at 1E6 cells per mL. Supernatants were collected and tested for human TGFβ1 using an ELISA-based kit (R&D Systems). The values in the bar graph represent the average TGFβ1 levels (pg/mL ± SD) from three biological replicates. (B) In vivo tumor growth control of ADI-270 and unarmored anti-CD70 CAR Vδ1 T cells against 4e6 SC-implanted A498 tumors in NSG mice. Mice were treated when the tumor size reached approximately 200 mm³ (Day 7 post-implantation). Data shown are the mean tumor volume ± SEM from five mice per group and represent two independent experiments. One-way ANOVA, ****P<0.0001, *p<0.05. (C) In vivo efficacy of a single IV dose of ADI-270 (1.6e7 cells, 52 days post-tumor engraftment) in R/R AML PDX model using NSG-SGM3 mice (J000106569, Jackson Lab). The expression of CD70⁺CD33⁺ blasts and % blasts (CD45⁺) from mouse blood at day-1 prior to treatment were performed using flow cytometry. Kaplan Meier analysis was used to show the probability of survival in each cohort (n=7 per group). ***p<0.001 (between ADI-270 and unarmored control), Log-rank (Mantel-Cox) test. ANOVA, analysis of variance; SC, subcutaneously; IV, intravenous; NSG-SGM3, NOD.Cg-Prkdc^{scid} IL2rg^{tm1Wjl} Tg(CMV-IL3,CSF2,KITLG)1Eav; PDX, patient-derived xenograft; R/R, relapsed/refractory; AML, acute myeloid leukemia.



Supplemental Fig. 8. Expression of the anti-CD70 CAR does not negatively impact manufacturing cell expansions. The fold expansions of Vδ1 T cells engineered to express the anti-CD70 CAR (blue circle) were compared to those expressing a non-CD70 targeting CAR (green square) at the end of the manufacturing process. The data points (symbols) are from multiple expansions from four different matching donors. The horizontal line indicates the average fold expansion of Vδ1 T cells. No statistically significant differences were found between the conditions (unpaired t-test).

Target	Clone	Catalog #	Supplier
Vδ1	REA173/R9.12	130-117-697	Miltenyi Biotech
Vδ2	B6	331432	Biolegend
TCRγδ	REA591/11F2 IMMU510	130-128-178 IM1571U	Miltenyi Biotech Beckman Coulter
TGFβRII	REA903/MM0056-4F14 25508	130-115-023 FAB241A-100	Miltenyi Biotech R&D Systems
CD3	UCHT1	A66327	Beckman Coulter
CD14	M5E2	301814	Biolegend
CD19	H1B19	302242	Biolegend
CD33	WM53	303408	Biolegend
CD56	HCD56	318328	Biolegend
TCRαβ	BW242/412	170-081-042	Miltenyi Biotech
CD62L	DREG-56	304818	Biolegend
CD95	DX2	305628	Biolegend
CD45	HI30	304028	Biolegend
CD45RO	UCHL1	304234	Biolegend
CD45RA	HI100	304128	Biolegend
CD70	113-16	355104	Biolegend
NKG2D	1D11	320824	Biolegend
DNAM1	11A8	338324	Biolegend
CCR5	J418F1	359110	Biolegend
CCR7	G043H7	353216	Biolegend
CXCR3	G025H7	353704	Biolegend
CXCR4	12G5	306510	Biolegend
4-1BB	4B4-1	741000	BD Bioscience
CD25	M-A251	356142	Biolegend
PD-1	NAT105	367404	Biolegend
TIM-3	F38-2E2	345022	Biolegend
LAG-3	11C3C65	369330	Biolegend
TIGIT	A15153G	372708	Biolegend
pSMAD2/3	O72-670	562586	BD Bioscience
P2A	2F13	ZRB1746	Sigma-Aldrich

Supplemental Table 1. Antibodies for flow cytometry studies

DEGs found in unarmored anti-CD70 CAR		DEGs shared in unarmored anti-CD70 CAR and ADI-270		DEGs found in ADI-270	
Upregulated	Downregulated	Upregulated	Downregulated	Upregulated	Downregulated
CXCL13 CCR4 CXCR4 AFDN TYK2 ATG14 CD9 TRBV28 JUN CTSD	ID2 PECAM1 SERPINB9 BCL6 SLAMF7 PTGER4 SMAD3 NCR3 OAS3 ITGAM BCL2L1 CD28 TCF7 CCL1 SOCS2 IL2RA TNF ITGB2 LTA IL12RB2 LAG3 TBX21 EBI3 PTGER2 ICAM1 CCR8 CSF2	CXCL13		CXCL13	

Supplemental Table 2. Differentially expressed genes (DEGs) in unarmored anti-CD70 CAR and ADI-270 following stimulation with recombinant CD70 protein in the presence of TGFβ1 compared to the absence of TGFβ1.